

GNN Coding Competition Template – Designer Manual

This document explains how to use the GNN Coding Competition Template ZIP folder to run a human-vs-LLM competition with automatic scoring and leaderboard updates on GitHub.

1. Purpose of the Template

The template is designed to evaluate coding and modeling capability in GNN competitions without executing untrusted participant code. Participants submit prediction files only, which are scored automatically using a fixed evaluation pipeline.

2. Contents of the ZIP Folder

After extracting the ZIP file, you will see the following main directories:

- data/: public datasets and sample submission
- competition/: validation and evaluation scripts
- submissions/: location where participant submissions are added
- leaderboard/: leaderboard files
- .github/workflows/: GitHub Actions for scoring and leaderboard updates
- README.md: competition description and rules

3. Uploading to GitHub

1. Create a new repository on GitHub.
2. Upload all extracted files to the repository root.
3. Commit and push the files to the main branch.
4. Enable GitHub Actions in the repository settings.

4. Preparing the Data

Public data goes into data/public/. This includes training, validation, and test inputs.

The true labels for the test set **must NOT** be committed to the repository. They should be stored privately and injected during CI evaluation using GitHub Secrets.

5. Submission Format

Each submission must include a predictions.csv file with the following columns:

- id: node or instance identifier
- y_pred: predicted probability or score

IDs must exactly match those in data/public/test_nodes.csv.

6. How Participants Submit

Participants submit by opening a Pull Request that adds:

submissions/inbox/<team_name>/<run_id>/predictions.csv

submissions/inbox/<team_name>/<run_id>/metadata.json

The metadata file records whether the submission was produced by a human, an LLM, or both.

7. Automatic Scoring

When a Pull Request is opened, GitHub Actions:

- Validates the submission format
- Scores predictions using hidden test labels
- Posts the score as a comment on the Pull Request

Invalid submissions fail automatically.

8. Leaderboard Updates

Once a submission Pull Request is merged, the leaderboard is updated automatically.

Scores are appended to leaderboard/leaderboard.csv and rendered in leaderboard.md.

9. Using the Template for Research

For human-vs-LLM studies, enforce identical constraints:

- Same time budget
- Same submission limit
- Same public data and evaluation rules

Recommended reported metrics include validity rate, best score, and score vs submission index.

10. Common Pitfalls

- Accidentally committing test labels
- Allowing participants to modify evaluation scripts
- Inconsistent ID matching between test nodes and predictions
- Changing metrics mid-competition

11. Customization

You may customize:

- Evaluation metric (competition/metrics.py)
- Task type (binary, multiclass, link prediction)
- Leaderboard display

Avoid changing rules after submissions begin.

12. Support

If you extend this template for academic use, document all changes clearly to preserve reproducibility.

13. Interactive Leaderboard (GitHub Pages)

The template includes an interactive leaderboard inspired by modern benchmark websites (e.g., search, filters, sortable columns, and column toggles). This leaderboard is implemented using **static HTML/CSS/JavaScript** and hosted via GitHub Pages.

Files involved:

- docs/leaderboard.html: main leaderboard page
- docs/leaderboard.css: styling
- docs/leaderboard.js: client-side logic
- leaderboard/leaderboard.csv: authoritative score data
- leaderboard/leaderboard.md: auto-generated static view for GitHub

13.1 Enabling GitHub Pages

To enable the interactive leaderboard:

1. Go to the repository Settings page.
2. Navigate to Pages.
3. Set Source to the main branch and folder to /docs.
4. Save the settings.

The leaderboard will then be available at:

<https://<your-organization>.github.io/<repository>/leaderboard.html>

13.2 How the Leaderboard Updates

The leaderboard update process is fully automated:

- When a submission Pull Request is opened, it is validated and scored.
- When the Pull Request is merged, the score is appended to leaderboard/leaderboard.csv.
- A GitHub Actions workflow runs competition/render_leaderboard.py to regenerate leaderboard.md.
- The interactive leaderboard automatically reflects the updated CSV without redeployment.

13.3 Filtering and Comparison Features

The interactive leaderboard supports:

- Free-text search over team, model, notes, and date
- Filtering by model type (e.g., human, llm-only, human+llm)
- Date-range filtering (recent submissions)
- Sorting by any column (default: score)
- Toggling visibility of columns

These features make it easy to compare human and LLM submissions at scale.

14. Updated Best Practices

- Treat leaderboard/leaderboard.csv as the single source of truth.
- Do not manually edit leaderboard.md; it is auto-generated.

- Do not commit private test labels.
- Announce any leaderboard rule changes clearly and before new submissions.

15. Verifying the Interactive Leaderboard Files

The ZIP template now explicitly includes the interactive leaderboard files. After extracting the ZIP, you should verify that the following files exist:

```
docs/leaderboard.html  
docs/leaderboard.css  
docs/leaderboard.js
```

If these files are present, the interactive leaderboard is correctly installed. These files are required for the GitHub Pages-based leaderboard and are separate from the static Markdown leaderboard.

15.1 Common Packaging Pitfall

If the interactive leaderboard does not appear after enabling GitHub Pages, the most common cause is uploading an outdated ZIP archive that does not include the docs/leaderboard.* files. Always ensure you are using the latest template ZIP.

15.2 Quick Verification Checklist

- docs/leaderboard.html exists
- docs/leaderboard.css exists
- docs/leaderboard.js exists
- GitHub Pages source is set to main /docs
- leaderboard/leaderboard.csv contains at least one valid entry

16. Summary of Leaderboard Architecture

The leaderboard system consists of two layers:

1. Authoritative data layer: leaderboard/leaderboard.csv
2. Presentation layer:
 - Static: leaderboard/leaderboard.md (auto-generated)
 - Interactive: docs/leaderboard.html (GitHub Pages)

Only the CSV should be treated as the source of truth.